ClassroomCast — Summary for the Teacher

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Meeting: 10 November 2026 · **Status:** Planning documents only. Nothing is built or tested yet.

1. What the project is

ClassroomCast is a local-network web system with three main functions:

1. **Screen sharing.** The teacher shares their screen and audio with 50–150 students. Students watch in a browser and install nothing.
2. **Class control.** The teacher creates a class with a join code and QR code, then approves, removes or blocks students.
3. **Website blocking from the dashboard.** An admin blocks AI sites, coding agents, social media and games using profiles such as Normal Class, Programming Lab and Exam Mode. A local DNS filter (AdGuard Home) enforces the rules, so nobody has to change router settings by hand.

It runs without internet, in two modes: on the **campus Wi-Fi** (with campus IT approval) or on a **teacher-owned router**.

2. What we uploaded (read in this order)

#	File	What it shows	Size
01	Project Proposal	Goal, scope, what is excluded, team hours (about 480 h)	234 lines
02	Requirements (SRS)	All 63 of your requirements (NAC, T, S, CN, OR) in one table	309 lines
03	Roles & Permissions	4 roles (Admin, Teacher, Student, Network Admin) and who can do what	90 lines
04	Feature Catalogue	68 feature cards grouped into 6 areas	793 lines
05	Use Cases & Flows	15 use cases (UC01–UC15) and 18 flow diagrams (F01–F18)	1,936 lines
06	System Design	Architecture, database design, API list	315 lines
07	Project Plan	Schedule: 3 sprints from Nov 30 to Jan 13, presentation Feb 3; task owners; risks	967 lines
08	Test Plan	63 test cases, one per requirement. Status: *Not run*	303 lines

To review quickly, read 01, 03 and the schedule table in 07. The other files hold the detail behind them.

3. How your requirements are covered

Your requirement section	Our file
1. Overview and target environment	01
2. Network blocking (NAC01–NAC20)	02, 04 (Area 03), 05 (F09–F15)
3. Broadcasting (T01–T11, S01–S10)	02, 04 (Areas 01–02), 05 (F04–F08)
4–5. Campus mode (CN01–CN10) and router mode (OR01–OR12)	02, 04 (Area 04), 05 (F02–F03)
6. Architecture (React, Node.js, SFU, SQLite, AdGuard, Caddy, Docker)	06

Priorities copied from your list: 31 Must, 7 Should and 3 Could. The 22 CN/OR items had no priority label, so we proposed treating them as Must.

4. The two folders

- **classroomcast_v1:** our first draft. It contains the requirements, architecture and the FigJam board images (feature cards, flow diagrams).
- **classroomCast_v2:** the cleaned-up, organised version. It reuses v1 content and fixes errors. Capacity is 50–150 students. Files 01–08 come from v2.

5. Proposed MVP scope (for your approval)

We have about 325 team hours for development and 3 sprints that include the holidays. To finish on time, we propose building a working minimum product (MVP) first, in **teacher-owned router mode**. We control that equipment, so we do not have to wait for campus IT.

Level	What it includes	Requirement IDs
A. Build and demo	Teacher creates a class, students join with a code or QR, teacher approves them, then shares screen and audio.	T01–T08, S01–S08
	Admin blocks categories and domains with profiles. Rules apply when class starts and restore when it ends, with status, audit log, emergency rollback and auto-expiry.	NAC01–08, NAC10–13, NAC16–20

Level	What it includes	Requirement IDs
	Teacher-owned router setup that works fully offline	OR02, OR03, OR05, OR07, OR11, OR12
B. Design and document	Campus mode: setup guide and IT checklist. We deploy it only if campus IT approves.	CN01–CN10
	Full 150-student test: we test with 10–20 real laptops, simulate the rest, and report both results separately.	CN06, OR01, OR04, OR08
	Hardware items: equipment list and setup guide	OR06, OR09, OR10
C. Only if time allows	Streaming diagnostics, file sharing, history, downloads, hand-raise/polls, scheduling, import/export, group policies	T09–T11, S09–S10, NAC09, NAC14, NAC15

Sprint goals:

- **Sprint 1 (Nov 30 – Dec 11):** create a class, join with a code or QR, approve students; 1 teacher shares their screen with 3 laptops on our router; AdGuard blocks one domain.
- **Sprint 2 (Dec 14 – 25):** audio, quality settings, pause/resume, remove/block, reconnect; policy profiles and categories; apply when class starts and restore when it ends.
- **Sprint 3 (Dec 28 – Jan 13):** emergency rollback, auto-expiry, audit log, local-traffic protection; test with 10–20 laptops; bug fixing only, no new features.

6. Questions for you

1. Do you approve the MVP scope above (A = build, B = document, C = optional)?
2. Is testing with 10–20 real laptops plus simulation acceptable for the 150-student requirement?
3. Is it acceptable to deliver campus mode as design and documentation only, unless campus IT approves DNS blocking?
4. Deadlines: is the final editing deadline **Jan 25**, with the **Jan 27** release and **Feb 3** presentation as separate dates?
5. Two documents list different task owners. Which one is correct?